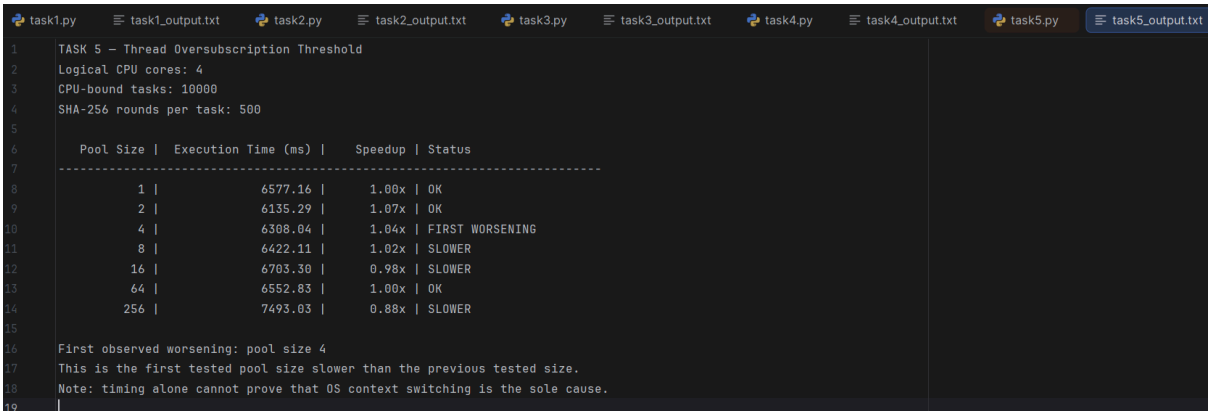

task1.py
task1_output.txt
task2.py
task2_output.txt
task3.py
task3_output.txt
task4.py
task4_output.txt
task5.py
task5_output.txt
TASK 5 — Thread Oversubscription Threshold
Logical CPU cores: 4
CPU-bound tasks: 10000
SHA-256 rounds per task: 500
Pool Size | Execution Time (ms) | Speedup | Status
1 | 6577.16 | 1.00x | OK
2 | 6135.29 | 1.07x | OK
4 | 6308.04 | 1.04x | FIRST WORSENING
8 | 6422.11 | 1.02x | SLOWER
16 | 6703.30 | 0.98x | SLOWER
64 | 6552.83 | 1.00x | OK
256 | 7493.03 | 0.88x | SLOWER
First observed worsening: pool size 4
This is the first tested pool size slower than the previous tested size.
Note: timing alone cannot prove that OS context switching is the sole cause.